

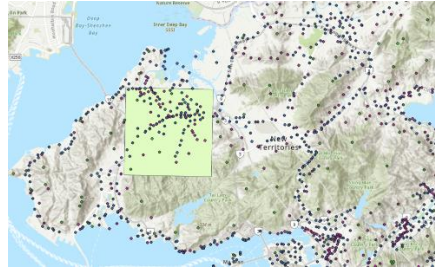
Data Terrain Modeling Lab 6 Assignment

1. Lab 6.1

1.1 Clipped Area of Interest



The survey data points of the trigonometrical stations, traverse stations, and benchmarks imported into ArcGIS

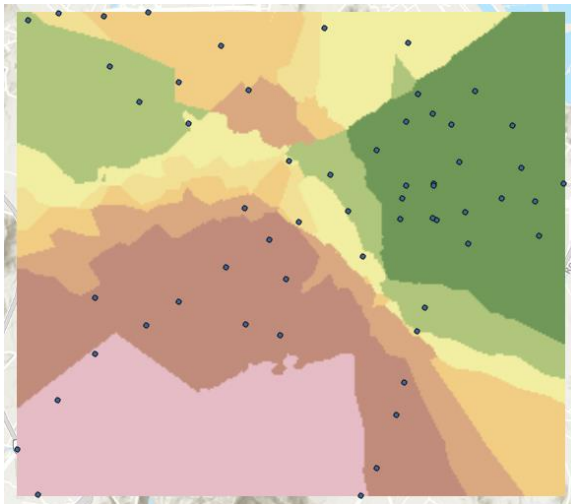


The area of interest to generate the DEM surface. In this assignment a section of Yuen Long is clipped to generate the DTM. It should contain more than 10 points.



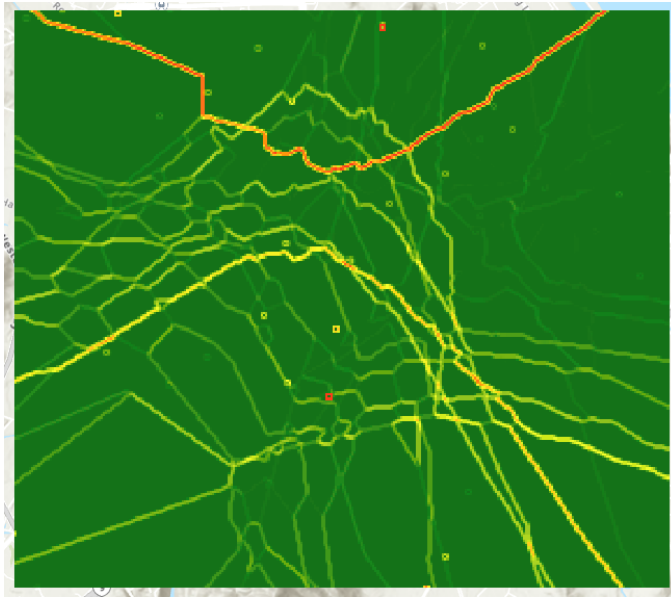
The final clipped area of interest and the included points.

1.2 Generated DEM



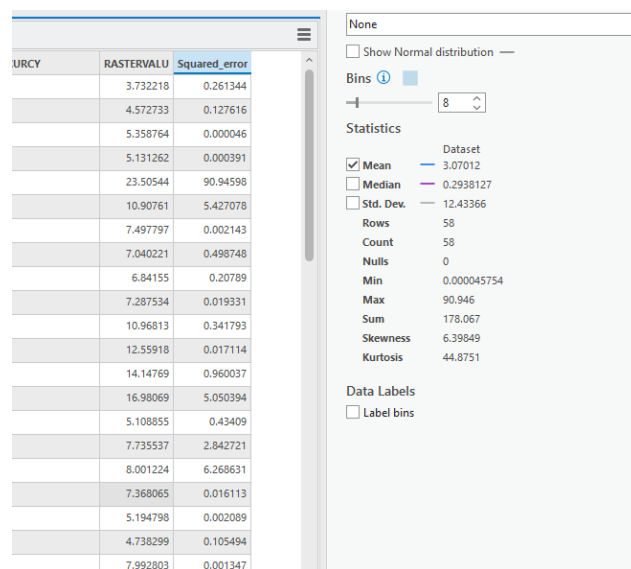
A DEM can be generated with the points inside the clipped area of interest. I chose the Kriging interpolation method for this assignment.

1.3 Slope Map



Generated slope map with “stretch” symbolization and “percent clip” stretch type

1.4 Calculation of the Accuracy Index



Screenshot of the squared error field and its statistics

Summary: A set of points are used to interpolate the DEM, then the points are extracted back to check the error against the original points.

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (Z_i^{DTM} - Z_i^{real})^2}{n}}$$

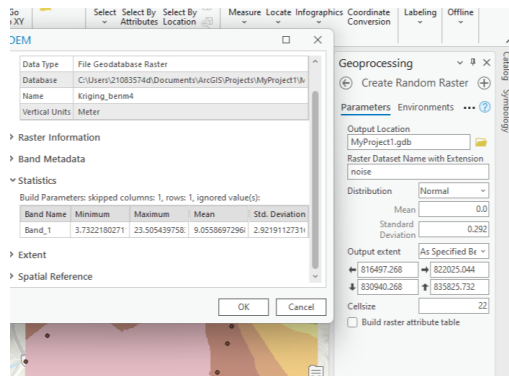
The formula for calculating the root mean square error (RMSE)

To calculate the RMSE, we first have to get the error for each field. This is done by calculating difference of the HKPD_m (elevation of the real point) and the RASTERVALU value (elevation of the interpolation) squared. We can store each squared error in the new

field and use the statistics to calculate the mean for us. In the above screenshot, the mean is 3.07012, which corresponds to the mean of the total squared errors. Therefore, we can calculate the accuracy index (RMSE) by taking its square root: $\sqrt{3.07012} = 1.752\text{m}$.

2. Lab 6.2

2.1 Generated Noise Raster

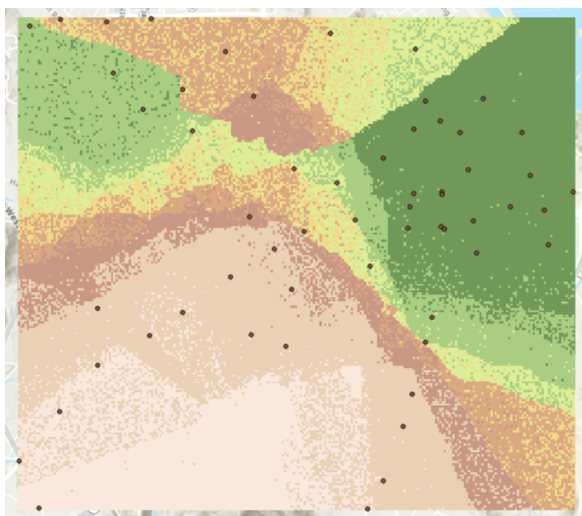


Using $1/10^{\text{th}}$ the standard deviation from the statistics, with mean 0.0 to generate a normally distributed random raster



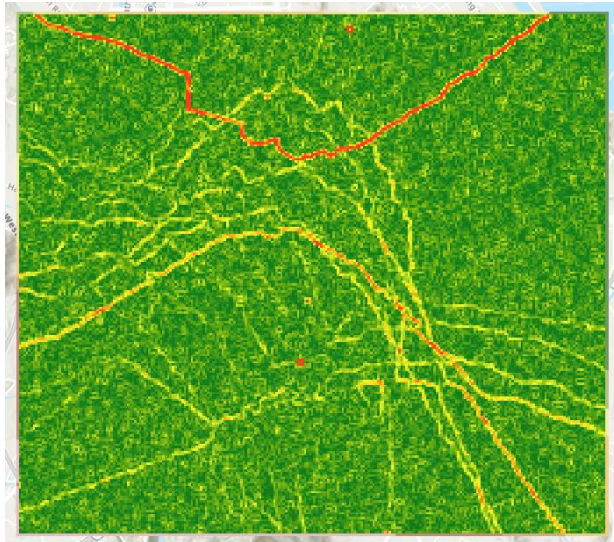
Generated random noise raster

2.2 DEM with noise



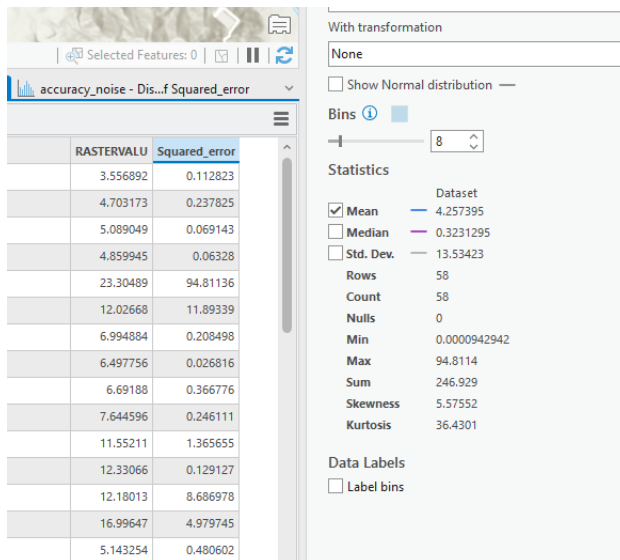
The new DEM after adding the noise raster to the original DEM raster using the Raster Calculator Tool. The points are obtained using the Extract Value from Points tool.

2.3 Slope map of DEM with noise



Slope Map with noise, using Percent Clip stretch style

2.4 Calculation of the Accuracy Index of DEM with noise



Screenshot of the squared error field and its statistics of the extracted points from DEM with noise

Similar to the previous task, the squared error is the squared difference of the elevations of the extracted point and the original elevation. We can use the statistics to find the mean is 4.257395, which corresponds to the mean of the sum of squared errors. Therefore the accuracy index (RMSE) is $\sqrt{4.257395} = 2.063\text{m}$.

3. Comparison of DEM accuracy with and without noise

Firstly, the original DEM surface appears to be smoother than the noise-added counterpart. Some pixels in the area where in the original DEM are the same color, have different colors in the DEM with noise. Random noise is the fluctuation of the data, therefore it should be expected that the DEM with random noise added appears rougher than the original DEM.

Secondly, in the original slope map, the slopes appear clearer and have higher contrast. However, the slopes in the slope map with noise added appears vaguer and have lower contrast. Though the main slopes in sharp red or yellow are still clearly visible, the slopes in lime or green are hard to see. This could be caused by the noise is obscuring part of the data, and the pixels blend in the original slopes.

Lastly, the RMSE of the extracted points in the DEM with noise is higher than which without noise. In the previous figures, the RMSE of the original DEM is 1.752m, and the RMSE of the noise-added DEM is 2.063m. This could be caused by the noise introduce some randomness to the original data, causing the data to more likely deviate from the real data, in other words the “errors” are more likely to occur. Since the RMSE takes into account of the squared errors, the noise is likely the cause of the increased RMSE.